WORKED SOLUTION

STEP II 2019, Question 9

Mechanics • Projectile Distance from Origin

STEP II 2019, Question 9

Read the question before you go on. It is in the STEP II paper in OCR's download of the 2019 STEP papers (a zip file):

www.ocr.org.uk/Images/696590-2019-past-papers.zip

This question has a full guided treatment in *Beyond STEP*, Section 11.10, *A Ball That Comes Back Towards You*: https://beyondstep.keithmcnulty.org/ch11/coming-back.html

The key idea

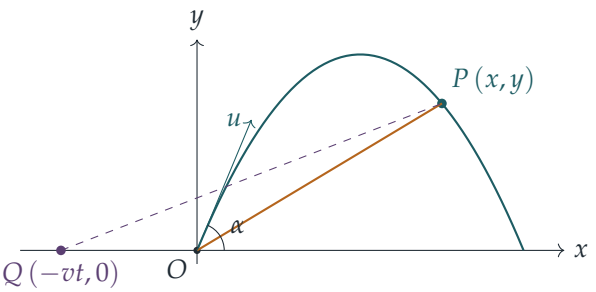
A distance is awkward to differentiate, but its square is not: OP is increasing exactly when $\frac{\mathrm{d}}{\mathrm{d}t}(OP^2) > 0$. Differentiating $x^2 + y^2$ gives t times a quadratic in t, so the whole question becomes "when is this quadratic positive?", which its discriminant answers. Part (ii) is the same calculation with Q's motion added to the horizontal separation.

Solution

Model P and Q as particles, ignore air resistance, and take O as the origin with x horizontal (in the direction of P's motion) and y vertically up. Write $s = \sin\alpha$ and $c = \cos\alpha$, both positive since $0 < \alpha < \frac{1}{2}\pi$. At time t,

$$x = uct, \qquad y = ust - \tfrac{1}{2}gt^2,$$

and P lands when $y = 0$ again, at $T = \dfrac{2us}{g}$.



Part (i)

Let $r = OP$, so $r^2 = x^2 + y^2$ and, using $\dot{x} = uc$ and $\dot{y} = us - gt$,

$$\begin{aligned}\frac{\mathrm{d}}{\mathrm{d}t}\left(r^2\right) = 2(x\dot{x} + y\dot{y}) &= 2\left[u^2c^2t + \left(ust - \tfrac{1}{2}gt^2\right)(us - gt)\right] \\ &= 2t\left[u^2c^2 + u^2s^2 - \tfrac{3}{2}usgt + \tfrac{1}{2}g^2t^2\right] \\ &= t\left(g^2t^2 - 3usg\,t + 2u^2\right),\end{aligned}$$

using $c^2 + s^2 = 1$. Since $r > 0$ during the flight, $\frac{\mathrm{d}}{\mathrm{d}t}(r^2) = 2r\dot{r}$ has the same sign as $\dot{r}$, and since $t > 0$ the sign is that of

$$f(t) = g^2t^2 - 3usg\,t + 2u^2.$$

This quadratic has positive leading coefficient and discriminant

$$9u^2s^2g^2 - 8u^2g^2 = u^2g^2\left(9s^2 - 8\right).$$

If $s < \frac{2\sqrt{2}}{3}$, then $9s^2 < 8$ (as $s > 0$), so the discriminant is negative and $f(t) > 0$ for every t. Hence $\dot{r} > 0$ throughout the flight: OP is increasing.

If $s > \frac{2\sqrt{2}}{3}$, the discriminant is positive and f has two distinct roots

$$t_{1,2} = \frac{u\left(3s \mp \sqrt{9s^2 - 8}\right)}{2g},$$

with $f(t) < 0$ for $t_1 < t < t_2$. Both roots are positive (their sum $3us/g$ and product $2u^2/g^2$ are positive). We must check that this interval actually occurs while P is in the air.

> **Ask yourself** The quadratic goes negative somewhere, but how do we know that happens before P lands?

Evaluate f at landing:

$$f(T) = g^2 \cdot \frac{4u^2s^2}{g^2} - 3usg \cdot \frac{2us}{g} + 2u^2 = 2u^2(1 - s^2) = 2u^2c^2 > 0.$$

So T is not between the roots. But the vertex of f is at $t = \frac{3us}{2g} < \frac{2us}{g} = T$, so T lies to the right of the vertex; being outside $[t_1, t_2]$, it must satisfy $T > t_2$. Hence $0 < t_1 < t_2 < T$, and for $t_1 < t < t_2$ we have $\dot{r} < 0$: OP is decreasing at some time before P lands, as required.

Part (ii)

Q slides on the smooth ground with constant speed v away from P's side, so at time t it is at $(-vt, 0)$. The horizontal separation of P and Q is $x + vt = wt$, where

$$w = uc + v > 0,$$

and the vertical separation is y. So $QP^2 = w^2t^2 + y^2$, and exactly as before

$$\begin{aligned}\frac{\mathrm{d}}{\mathrm{d}t}\left(QP^2\right) &= 2\left[w^2t + \left(ust - \tfrac{1}{2}gt^2\right)(us - gt)\right] \\ &= t\left(g^2t^2 - 3usg\,t + 2w^2 + 2u^2s^2\right).\end{aligned}$$

The sign during the flight is that of $h(t) = g^2t^2 - 3usg\,t + 2(w^2 + u^2s^2)$, whose discriminant is

$$9u^2s^2g^2 - 8g^2\left(w^2 + u^2s^2\right) = g^2\left(u^2s^2 - 8w^2\right).$$

This is negative exactly when $8w^2 > u^2s^2$, that is (both w and us being positive) when $2\sqrt{2}\,w > us$. Substituting $w = uc + v$,

$$2\sqrt{2}\,(uc + v) > us \iff 2\sqrt{2}\,v > \left(\sin\alpha - 2\sqrt{2}\cos\alpha\right)u.$$

So under the given condition $h(t) > 0$ for all t, hence $\frac{\mathrm{d}}{\mathrm{d}t}(QP^2) > 0$ for $t > 0$ and QP is increasing throughout the flight of P, as required.

The same argument as in part (i) (now $h(T) = 2w^2 > 0$) shows that when the inequality is reversed strictly, QP does decrease for a while before P lands, so the condition is in fact also necessary, apart from the boundary case of equality.

Checks

- Setting $v = 0$ in part (ii) gives $2\sqrt{2}\cos\alpha > \sin\alpha$, i.e. $\tan\alpha < 2\sqrt{2}$, i.e. $\sin\alpha < \frac{2\sqrt{2}}{3}$: part (i) again, as it should be since Q then stays at O.
- The critical angle is $\arcsin\frac{2\sqrt{2}}{3} \approx 70.5^\circ$, independent of u and g, as dimensional analysis demands (the only dimensionless quantity available is α).
- With $u = 10\,\mathrm{m\,s^{-1}}$, $g = 9.8\,\mathrm{m\,s^{-2}}$ and $\sin\alpha = 0.97$, the roots are $t_1 \approx 1.14\,\mathrm{s}$ and $t_2 \approx 1.83\,\mathrm{s}$, both before landing at $T \approx 1.98\,\mathrm{s}$. Stepping through the flight by computer confirms that OP falls between these times, and that across many random u, α, v the distance QP is increasing exactly when the part (ii) inequality holds.

Common slips

- Stopping at "the discriminant is positive" in part (i): you must show the dip in OP happens before P lands, not after.
- Putting Q at $(vt, 0)$: Q moves away from P's side, so its horizontal separation from P is $uct + vt$.
- Forgetting the factor t, or dividing by it without noting $t > 0$.

Beyond STEP

Studying the sign of $\frac{\mathrm{d}}{\mathrm{d}t}|\mathbf{r}|^2 = 2\,\mathbf{r}\cdot\dot{\mathbf{r}}$ is the standard way to ask whether a body is approaching or receding, and it reappears in orbital mechanics and in the relative motion of two bodies, where part (ii) is really a question about the position of P relative to Q. The neat fact that a projectile thrown more steeply than about 70.5° comes back towards its thrower for part of its flight is a favourite of mechanics texts. For more, see Kleppner and Kolenkow, *An Introduction to Mechanics*.

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